Phys 115A HW 5

Zih-Yu Hsieh

November 21, 2025

1

Question 1. *Odd Wells (Griffiths 2.29)*
Let's go back to the odd case of the finite square well:
1. Write down the general form of the odd bound-state wavefunctions in the three regions $x < -a$, $-a \leq x \leq a$, and $x > a$.
2. Find the transcendental equation for the allowed energies. Solve it graphically.
3. What is the energy spectrum of bound states in the limiting cases of a wide, deep well and a shallow, narrow well (in the sense that Griffiths describes in Section 2.6 around Eq. 2.157)? Do we always find an odd bound state?

Proof. Recall that the general solution to the bound-state wavefunction of the symmetric finite square well (given that the potential is $-V_0$ on $[-a, a]$ for some $V_0 > 0$, and 0 elsewhere), is given as follow:

$$\psi(x) = \begin{cases} Ae^{kx} & x < -a \\ C\sin(\ell x) + D\cos(\ell x) & x \in [-a, a] \\ Be^{-kx} & x > a \end{cases} \tag{1}$$

Where the coefficient $k = \frac{\sqrt{-2mE}}{\hbar}$, and $\ell = \frac{\sqrt{2m(E+V_0)}}{\hbar}$.

1. The general form of the odd bound-state wavefunction is then given as follow:

$$\psi_{\text{odd}}(x) = \begin{cases} -Ae^{kx} & x < -a \\ C\sin(\ell x) & x \in [-a, a] \\ Ae^{-kx} & x > a \end{cases} \tag{2}$$

   Where the coefficient k, ℓ are listed above.

2. Using continuity and differentiability, we get the following implication:

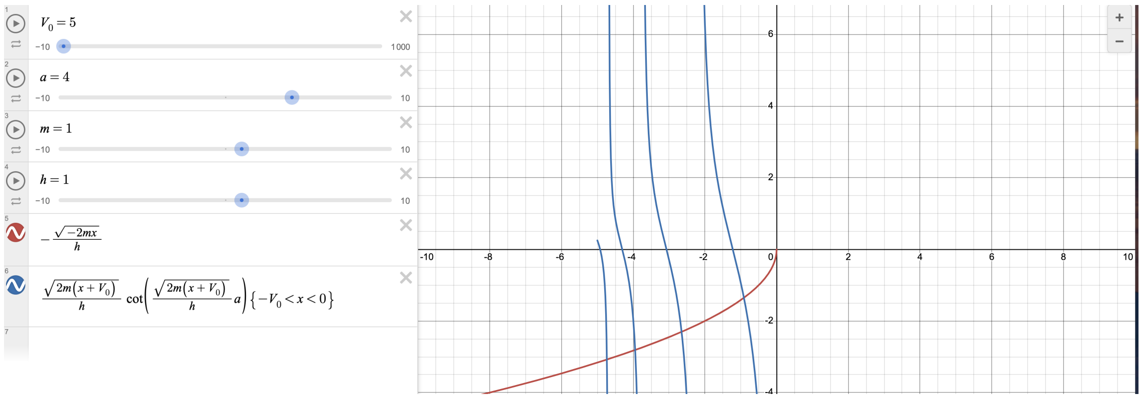
$$\text{continuity} \implies -Ae^{k(-a)} = C\sin(\ell(-a)) \implies Ae^{k(-a)} = C\sin(\ell a) \tag{3}$$

$$\text{differentiability} \implies -kAe^{k(-a)} = \ell C\cos(\ell(-a)) = \ell C\cos(\ell a) \tag{4}$$

Dividing the equations, we get that $-k = \ell \cot(\ell a)$. Which, plug in k, ℓ as their original expression, we get:

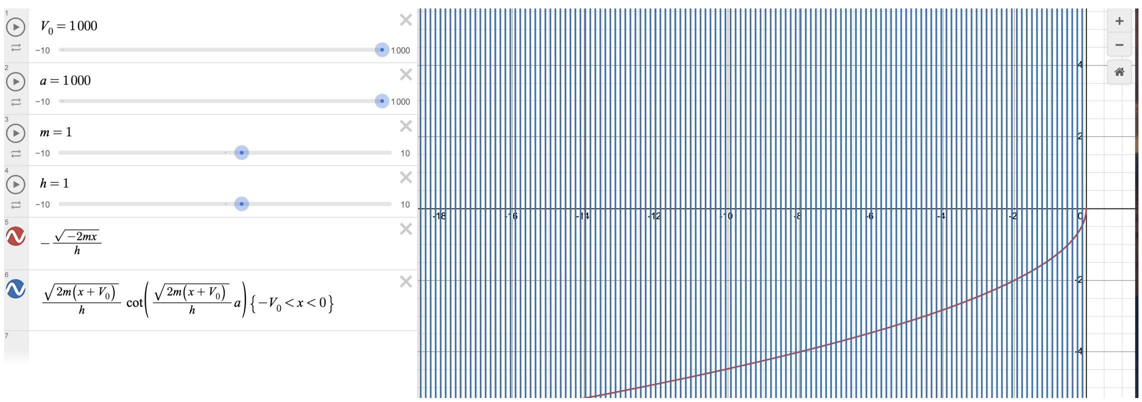
$$-\frac{\sqrt{-2mE}}{\hbar} = \frac{\sqrt{2m(E+V_0)}}{\hbar} \cot\left(\frac{\sqrt{2m(E+V_0)}}{\hbar} a\right) \tag{5}$$

Fixing constant $V_0, a, m, \hbar > 0$, we get the following graph: Where, the intersection points are the x



(or E in the previous equations) that provides the bound-state energy.

3. If the well is wide and deep, then graphically it is verified that the energy spectrum will range widely, and becomes more dense; graphically, here is an example: on the other hand, a shallow and narrow



one (an example would be the first graph provided), has only few energy spectrum allowed (and are pretty limited).

In this case it's not guaranteed to find an odd bound state: Graphically when we plot $a = \frac{1}{V_0}$ for instance, after a certain value the two equations no longer intersect, indicating there's no allowed odd bound-state (since there is no corresponding energy allowed based on calculation).

$\square$

2

Question 2. *Delta limit (Griffiths 2.31)*
As we discussed in class, a Dirac delta function can be thought of as a limiting case of a finite square well, with the added condition that the area is fixed to be 1 as the height goes to infinity and the width goes to zero. Show that the delta function well is a "weak" potential in the sense that $z_0 \to 0$. *Determine the bound state energy for the delta-function potential by treating it as the limit of the finite square well. Check that your answer is consistent with Eq.* 2.132. *Also show that Eq.* 2.172 *reduces to Eq.* 2.144 *in the appropriate limit.*

Proof. Given the condition that the area of the well is fixed to 1, then with the (symmetric) well of width $2a$ and depth V_0, we need $2aV_0 = 1$, so fixing V_0, we get $a = \frac{1}{2V_0}$. As a result, the equation $z_0 := \frac{a}{\hbar}\sqrt{2mV_0}$ satisfies the following:

$$\lim_{V_0\to\infty} z_0 = \lim_{V_0\to\infty} \frac{\sqrt{2m}}{2\hbar V_0} \cdot \sqrt{V_0} = \lim_{V_0\to\infty} \frac{\sqrt{2m}}{2\hbar\sqrt{V_0}} = 0 \tag{6}$$

Since it's guaranteed that at least one of the bound-state is left behind in the well (regardless of the well's shape), and such left behind bound-state is always given by an even bound-state, which based on the even bound-state energy equation, we get the following:

$$\frac{\sqrt{-2mE}}{\hbar} = \frac{\sqrt{2m(E+V_0)}}{\hbar} \tan\left(\frac{\sqrt{2m(E+V_0)}}{\hbar} a\right) = \frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar} \cdot \sqrt{V_0} \tan\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar} \frac{\sqrt{V_0}}{2V_0}\right) \tag{7}$$

$$= \frac{2m(\frac{E}{V_0}+1)}{\hbar^2} \frac{\tan\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar}\frac{1}{2\sqrt{V_0}}\right)}{\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar}\frac{1}{2\sqrt{V_0}}} = \frac{2m(\frac{E}{V_0}+1)}{\hbar^2} \frac{\tan\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar}\frac{1}{2\sqrt{V_0}}\right)}{\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar}\frac{1}{2\sqrt{V_0}}} \tag{8}$$

Which, let $y := \frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar}\frac{1}{2\sqrt{V_0}}$, we have $\lim_{V_0\to\infty} y = 0$. So, we get the following:

$$\lim_{V_0\to\infty} \frac{\sqrt{2mE}}{\hbar} = \lim_{V_0\to\infty} \frac{2m(\frac{E}{V_0}+1)}{2\hbar^2} \frac{\tan(y)}{y} = \frac{2m}{2\hbar^2} = \frac{m}{\hbar^2} \tag{9}$$

Hence, under the limit of delta square well, $\frac{\sqrt{-2mE}}{\hbar} = \frac{m}{\hbar^2}$, so $-2mE = \frac{m^2}{\hbar^2}$, showing $E = -\frac{m}{2\hbar^2}$. Which, given that the delta well has area 1, then it's given by $V(x) = -\delta(x)$ (so the coefficient $\alpha = -1$), then Eq. 2.132 provides the energy $E = -\frac{m\alpha^2}{2\hbar^2} = -\frac{m}{2\hbar^2}$ agrees with the limit.

Finally, given Eq. 2.172 as follow:

$$T^{-1} = 1 + \frac{V_0^2}{4E(E+V_0)} \sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right) \tag{10}$$

Also, Eq. 2.144 is as follow:

$$T = \frac{1}{1 + (m\alpha^2/2\hbar^2 E)} \tag{11}$$

Which, according to Eq. 2.172, it can be rewritten as follow (based on the fact that $a = \frac{1}{2V_0}$ under our assumption):

$$T^{-1} = 1 + \frac{V_0}{4E(\frac{E}{V_0}+1)}\sin^2\left(\frac{1}{\hbar V_0}\sqrt{2m\left(\frac{E}{V_0}+1\right)}\sqrt{V_0}\right) = 1 + \frac{1}{4E}\left(\frac{\sin\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}}\right)}{\sqrt{\frac{E}{V_0}+1}\frac{1}{\sqrt{V_0}}}\right)^2 \tag{12}$$

$$= 1 + \frac{2m}{\hbar^2 4E}\left(\frac{\sin\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}}\right)}{\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}}}\right)^2 \tag{13}$$

Hence, we get the following limit, due to the fact that $\lim_{V_0\to\infty}\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}} = 0$:

$$\lim_{V_0\to\infty} T^{-1} = \lim_{V_0\to\infty} 1 + \frac{2m}{\hbar^2 4E}\left(\frac{\sin\left(\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}}\right)}{\frac{\sqrt{2m(\frac{E}{V_0}+1)}}{\hbar\sqrt{V_0}}}\right)^2 = 1 + \frac{2m}{\hbar^2 4E} \tag{14}$$

Hence, under the limit, we have $T = \frac{1}{T^{-1}} = \frac{1}{1+2m/(\hbar^2 4E)} = \frac{1}{1+m/(2\hbar^2 E)}$. Which, if cf. Eq. 2.144, with $\alpha = -1$ (based on what we previously get), $T = \frac{1}{1+m\alpha^2/(2\hbar^2 E)} = \frac{1}{1+m/(2\hbar^2 E)}$, which coincides with our limit. $\square$

3

Question 3. *The Square Barrier (Griffiths 2.33)*
Let's work out the transmission coefficient for a rectangular barrier defined by
$V(x) = \{V_0 \text{ for } -a \leq x \leq a,\ 0 \text{ for } |x| > a\}$
for $V_0 > 0$. *There are three possible cases of interest:* $E < V_0$; $E = V_0$; *and* $E > V_0$, *and the wavefunction inside the barrier is different in each case.*
1. Find T^{-1} *for* $E < V_0$
2. Find T^{-1} *for* $E = V_0$
3. Find T^{-1} *for* $E > V_0$ *(Hint: This one is easily related to the finite square well.)*
Partial answer: for $E < V_0$, $T^{-1} = 1 + \frac{V_0^2}{4E(V_0-E)} \sinh^2\left(\frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right)$

Proof. (**Rmk:** The calculation is too tedious to include every detail, we'll just summarize when necessary).

Regardless of the case, on the intervals $(-\infty, a)$ and (a, ∞), since $V(x) = 0$, the time-independent Schrödinger equation (TISE) is given by $E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}$, which let $k := \frac{\sqrt{2mE}}{\hbar}$, the equation becomes $\frac{\partial^2\psi}{\partial x^2} + k^2\psi = 0$, so $\psi(x) = Ae^{ikx} + Be^{-ikx}$. WLOG assume that the source of the particles are only from the left to the right, then on the two regions, we have the following solution for $A, B, F \in \mathbb{C}$:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x \in (-\infty, -a) \\ Fe^{ikx} & x \in (a, \infty) \end{cases} \tag{15}$$

1. If given that $E < V_0$, then on the region $(-a, a)$, the TISE becomes the following:

$$E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V_0\psi \implies \frac{\partial^2\psi}{\partial x^2} - \frac{2m(V_0 - E)}{\hbar^2}\psi = 0 \tag{16}$$

Which, with $w := \frac{\sqrt{2m(V_0-E)}}{\hbar}$, on the region $(-a, a)$ the solution is $\psi(x) = Ce^{wx} + De^{-wx}$ for $C, D \in \mathbb{C}$.

Which, using both continuity and differentiability at $\pm a$, we get the following four equations:

$$\begin{cases} Ae^{-ika} + Be^{ika} = Ce^{-wa} + De^{wa} \\ ikAe^{-ika} - ikBe^{ika} = wCe^{-wa} - wDe^{wa} \\ Ce^{wa} + De^{-wa} = Fe^{ika} \\ wCe^{wa} - wDe^{-wa} = ikFe^{ika} \end{cases} \tag{17}$$

Using the first two equations, C, D can be expressed using the following matrix equation:

$$\begin{pmatrix} e^{-wa} & e^{wa} \\ we^{-wa} & -we^{wa} \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} Ae^{-ika} + Be^{ika} \\ ikAe^{-ika} - ikBe^{ika} \end{pmatrix} \tag{18}$$

Which, after solving the linear equations, we get the following:

$$C = \frac{e^{wa}}{2}\left(\left(1 + \frac{ik}{w}\right)Ae^{-ika} + \left(1 - \frac{ik}{w}\right)Be^{ika}\right) \tag{19}$$

$$D = \frac{e^{-wa}}{2}\left(\left(1 - \frac{ik}{w}\right)Ae^{-ika} + \left(1 + \frac{ik}{w}\right)Be^{ika}\right) \tag{20}$$

Which, this implies the following:

$$Ce^{wa} = \frac{e^{2wa}}{2}\left(\left(1+\frac{ik}{w}\right)Ae^{-ika} + \left(1-\frac{ik}{w}\right)Be^{ika}\right) \tag{21}$$

$$De^{-wa} = \frac{e^{-2wa}}{2}\left(\left(1-\frac{ik}{w}\right)Ae^{-ika} + \left(1+\frac{ik}{w}\right)Be^{ika}\right) \tag{22}$$

Now, the third and the fourth equation provides the following equality:

$$Ce^{wa} + De^{-wa} = Fe^{ika} = -\frac{iw}{k}Ce^{wa} + \frac{iw}{k}De^{-wa} \implies \left(1+\frac{iw}{k}\right)Ce^{wa} + \left(1-\frac{iw}{k}\right)De^{-wa} = 0 \tag{23}$$

Plugin the previous equalities about Ce^{wa}, De^{-wa} (with some simplification), we get:

$$0 = \left(1+\frac{iw}{k}\right)\frac{e^{2wa}}{2}\left(\left(1+\frac{ik}{w}\right)Ae^{-ika}\left(1-\frac{ik}{w}\right)Be^{ika}\right) \tag{24}$$

$$+\left(1-\frac{iw}{k}\right)\frac{e^{-2wa}}{2}\left(\left(1-\frac{ik}{w}\right)Ae^{-ka} + \left(1+\frac{ik}{w}\right)Be^{ika}\right) \tag{25}$$

$$\implies A\left(\frac{w^2+k^2}{kw}\right)ie^{-ika}\sinh(2wa) = -Be^{ika}\left(2\cosh(2wa) + i\frac{w^2-k^2}{kw}\sinh(2wa)\right) \tag{26}$$

$$\implies |A|^2\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa) = |B|^2\left(4+\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa)\right) \tag{27}$$

$$\implies \frac{|B|^2}{|A|^2} = \frac{\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa)}{4+\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa)} \tag{28}$$

Which, the transmission rate $T = 1-\frac{|B|^2}{|A|^2}$ can be given as below:

$$T = 1-\frac{|B|^2}{|A|^2} = \frac{4}{4+\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa)} \tag{29}$$

$$\implies T^{-1} = 1+\frac{1}{4}\left(\frac{w^2+k^2}{wk}\right)^2\sinh^2(2wa) \tag{30}$$

Finally, recall that $w = \frac{\sqrt{2m(V_0-E)}}{\hbar}$ and $k = \frac{\sqrt{2mE}}{\hbar}$, so $w^2+k^2 = \frac{2m(V_0-E)}{\hbar^2} + \frac{2mE}{\hbar^2} = \frac{2mV_0}{\hbar^2}$, while $kw = \frac{2m\sqrt{E(V_0-E)}}{\hbar^2}$. This shows that $\left(\frac{w^2+k^2}{wk}\right)^2 = \frac{V_0^2}{E(V_0-E)}$. Hence, T^{-1} expands to be the following:

$$T^{-1} = 1+\frac{1}{4}\cdot\frac{V_0^2}{E(V_0-E)}\sinh^2\left(2a\cdot\frac{\sqrt{2m(V_0-E)}}{\hbar}\right) = 1+\frac{V_0^2}{4E(V_0-E)}\sinh^2\left(\frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right) \tag{31}$$

2. If given that $E = V_0$, then on the region $(-a, a)$, the TISE becomes the following:

$$V_0\psi = E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V_0\psi \implies \frac{\partial^2\psi}{\partial x^2} = 0 \tag{32}$$

Hence, on the region $(-a, a)$, the solution is $\psi(x) = Cx + D$ for $C, D \in \mathbb{C}$.

Together with the solutions outside of this region, based on the continuity of both ψ and $\frac{d\psi}{dx}$ at $-a$ and a, we get the following four equations:

$$\begin{cases} Ae^{-ika} + Be^{ika} = -Ca + D \\ ikAe^{-ika} - ikBe^{ika} = C \\ Ca + D = Fe^{ika} \\ C = ikFe^{ika} \end{cases} \tag{33}$$

The first two equations can be collected into the following linear system:

$$\begin{pmatrix} e^{-ika} & e^{ika} \\ ike^{-ika} & -ike^{ika} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} -Ca + D \\ C \end{pmatrix} \tag{34}$$

Which, after solving, we get that $A = De^{ika}$, and $B = -Cae^{-ika}$. Now, use $D = Ae^{-ika}$ and $Ca = -Be^{ika}$, plugin to the third and the fourth equation, we get:

$$\begin{cases} Fe^{ika} = -Be^{ika} + Ae^{-ika} \\ ikFe^{ika} = -\frac{Be^{ika}}{a} \end{cases} \implies Ae^{-ika} - Be^{ika} = -\frac{Be^{ika}}{ika} \tag{35}$$

$$\implies Ae^{-ika} = Be^{ika}\left(1 - \frac{1}{ika}\right) \tag{36}$$

$$\implies |A|^2 = |B|^2 \cdot \left|\frac{ika - 1}{ika}\right|^2 = \frac{1 + (ka)^2}{(ka)^2} \tag{37}$$

As a result, we get the reflection rate $R = \frac{|B|^2}{|A|^2} = \frac{(ka)^2}{1+(ka)^2}$, so the transmission rate $T = 1 - R = \frac{1}{1+(ka)^2}$. Hence, T^{-1} can be expressed as follow, given $E = V_0$, and $k = \frac{\sqrt{2mE}}{\hbar}$:

$$E = V_0 \implies T^{-1} = 1 + (ka)^2 = 1 + \frac{2mEa^2}{\hbar^2} = 1 + \frac{2mV_0a^2}{\hbar^2} \tag{38}$$

3. If given that $E > V_0$, then on the region $(-a, a)$, the TISE becomes the following:

$$E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V_0\psi \implies \frac{\partial^2\psi}{\partial x^2} + \frac{2m(E - V_0)}{\hbar^2}\psi = 0 \tag{39}$$

Which, with $w := \frac{\sqrt{2m(E-V_0)}}{\hbar}$, on the region $(-a, a)$ the solution is $\psi(x) = Ce^{iwx} + De^{-iwx}$ for $C, D \in \mathbb{C}$.

Then, with both continuity and differentiability at $\pm a$, we get the following four equations:

$$Ae^{-ika} + Be^{ika} = Ce^{-ika} + De^{ika} \tag{40}$$

$$ikAe^{-ika} - ikBe^{ika} = iwCe^{-ika} - iwDe^{iwa} \tag{41}$$

$$Ce^{iwa} + De^{-iwa} = Fe^{ika} \tag{42}$$

$$iwCe^{iwa} - iwDe^{-iwa} = ikFe^{ika} \tag{43}$$

Notice that this relation is similar to the finite square well (basically it's with the energy being V_0 instead of $-V_0$, since $E > V_0$, it's similar to the finite square well case where $E > -V_0$). Using the equation from finite square well, with the modification of $-V_0$ to V_0, we get:

$$T^{-1} = 1 + \frac{V_0^2}{4E(E - V_0)}\sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E - V_0)}\right) \tag{44}$$

$\square$

4

Question 4. *Mass on (circular) string (Griffiths 2.46)*
Imagine a bead of mass m that slides frictionlessly around a circular wire ring of circumference L. (This is like a free particle except that $\psi(x+L) = \psi(x)$.) Find the stationary states (with appropriate normalization) and the corresponding allowed energies. Note that there are (with one exception) two independent solutions for each energy E_n—corresponding to clockwise and counter-clockwise circulation; call them $\psi_n^+(x)$ and $\psi_n^-(x)$. How do you account for this degeneracy, in view of the theorem of problem 2.44 (why does the theorem fail in this case)? Hint: you worked through problem 2.44 in discussion section.

Proof. Given that $V(x) = 0$ where $x \in \mathbb{R}/(L\mathbb{Z})$ (i.e. it's periodic with period L position wise). Then, the Hamiltonian is given by $\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$. So, the stationary state satisfies the following equation with some energy $E \geq 0$:

$$E\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} \implies \frac{\partial^2\psi}{\partial x^2} + \frac{2mE}{\hbar^2}\psi = 0 \implies \psi(x) = Ae^{ikx} + Be^{-ikx}, \quad k := \frac{\sqrt{2mE}}{\hbar} \tag{45}$$

Which, using the periodicity, since $\psi(x+L) = \psi(x)$, one must have the following:

$$Ae^{ikx} + Be^{-ikx} = \psi(x) = \psi(x+L) = Ae^{ik(x+L)} + Be^{-ik(x+L)} = (Ae^{ikL})e^{ikx} + (B^{-ikL})e^{-ikx} \tag{46}$$

Then, by linear independence of the two functions e^{ikx}, e^{-ikx} over $\mathbb{C}$, it enforces $A = Ae^{ikL}$ and $B = Be^{-ikL}$ for all $A, B \in \mathbb{C}$, showing $e^{ikL} = e^{-ikL} = 1$, which $kL = 2n\pi$ for some $n \in \mathbb{Z}$, or $\frac{\sqrt{2mE}}{\hbar} = \frac{2n\pi}{L}$, showing that $E = \frac{1}{2m}\left(\frac{2n\pi\hbar}{L}\right)^2$ for some $n \in \mathbb{N}$ (since it's squared).

Fix $n \in \mathbb{N}$, define $E_n := \frac{2n^2\pi^2\hbar^2}{mL^2}$, then $k_n := \frac{\sqrt{2mE_n}}{\hbar} = \frac{2n\pi}{L}$. Let $\psi_n(x) := Ae^{ik_nx} + Be^{-ik_nx}$ be the corresponding n^{th} stationary state. To normalize this, since it's restricted to $\mathbb{R}/L\mathbb{Z}$, then the most fundamental domain is up to $[0, L)$. Then, to normalize the solution above, we get:

$$1 = \int_0^L |\psi|^2 dx = \int_0^L (Ae^{ik_nx} + Be^{-ik_nx})^*(Ae^{ik_nx} + Be^{-ik_nx})dx \tag{47}$$

$$= \int_0^L \left(|A|^2 + |B|^2 + A^*Be^{-i2k_nx} + AB^*e^{i2k_nx}\right)dx = L(|A|^2 + |B|^2) \tag{48}$$

(Note: Here e^{i2k_nx}, e^{-i2k_nx} still have period multiples being L).

Hence, one must have $|A|^2 + |B|^2 = \frac{1}{L}$. Hence, given any $0 \leq D_1 \leq \frac{1}{\sqrt{L}}$, with $D_2 := \sqrt{\frac{1}{L} - D_1^2} \geq 0$, given any $\theta_1, \theta_2 \in [0, 2\pi]$, the constants $A := D_1e^{i\theta_1}$ and $B := D_2e^{i\theta_2}$, they satisfy $|A|^2 + |B|^2 = |D_1|^2 + |D_2|^2 = D_1^2 + (\frac{1}{L} - D_1^2) = \frac{1}{L}$, which satisfies the given condition.

Which, incorporate for the time evolution (where fixing $n \in \mathbb{N}$, we have $E_n = \frac{2n^2\pi^2\hbar^2}{mL}$, $k_n = \frac{\sqrt{2mE_n}}{\hbar} = \frac{2n\pi}{L}$, and $\frac{iE_nt}{\hbar} = \frac{i\cdot 2n^2\pi^2\hbar}{mL^2}$), we get the following:

$$\psi_n(x,t) = A\exp\left(\frac{i2n\pi}{L}x - \frac{i2n^2\pi^2\hbar}{mL^2}t\right) + B\exp\left(-\frac{i2n\pi}{L}x - \frac{i2n^2\pi^2\hbar}{mL^2}t\right), \quad |A|^2 + |B|^2 = \frac{1}{L} \tag{49}$$

If restrict to $B = 0$ or $A = 0$ respectively, one can get $A := \frac{1}{\sqrt{L}}$ or $B := \frac{1}{\sqrt{L}}$, and one can define $\psi_n^+(x) := \frac{1}{\sqrt{L}}e^{\frac{i2n\pi}{L}x}$ and $\psi_n^-(x) := \frac{1}{\sqrt{L}}e^{-\frac{i2n\pi}{L}x}$, which they form a $\mathbb{C}$-linearly independent solutions, and form a basis for the stationary state for energy E_n.

The reason why theorem in Problem 2.44 fails here (where ψ_n^+ and ψ_n^- are non-degenerate), is because the the domain is bounded - instead of on the whole real line $x \in \mathbb{R}$, we have $x \in \mathbb{R}/(L\mathbb{Z})$ (which can essentially be described by $[0, L)$), since the domain is somewhat limited, it takes away extra restrictions (for instance $\lim_{x\to\infty} \psi(x) = 0$ is no longer required), hence the solution space essentiallly gets enlarged, and causing such degeneracy to be possible (since the proof in Problem 2.44 usees the fact that $\psi(x)$ converges to 0 when approaching $\pm\infty$). $\square$

5

Question 5. *The Transfer Matrix (Griffiths 2.54)*
The transfer matrix. The S-matrix (Problem 2.53) tells you the outgoing amplitudes (B and F) in terms of the incoming amplitudes (A and G) – Equation 2.180. For some purposes it is more convenient to work with the transfer matrix, M, which gives you the amplitudes to the right of the potential (F and G) in terms of those to the left (A and B):
$(F, G)^T = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} (A, B)^T$
a. Find the four elements of the M-matrix, in terms of the elements of the S-matrix, and vice versa. Express R_l, T_l, R_r, and T_r (Equations 2.176 and 2.177) in terms of elements of the M-matrix.
b. Suppose you have a potential consisting of two isolated pieces (Figure 2.23). Show that the M-matrix for the combination is the product $M = M_2M_1$.
c. Construct the M-matrix for scattering from a single delta-function potential at point a: $V(x) = -\alpha\delta(x - a)$
d. By the method of part (b), find the M-matrix for scattering from the double delta function $V(x) = -\alpha[\delta(x + a) + \delta(x - a)]$ What is the transmission coefficient for this potential?

Proof.

a. Given the relation with the S-matrix as follow:

$$\begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix} = \begin{pmatrix} AS_{11} + GS_{12} \\ AS_{21} + GS_{22} \end{pmatrix} \tag{50}$$

With the transfer matrix relation and the above S-matrix relation, we get:

$$\begin{pmatrix} AS_{21} + GS_{22} \\ G \end{pmatrix} = \begin{pmatrix} F \\ G \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} AM_{11} + BM_{12} \\ AM_{21} + BM_{22} \end{pmatrix} = \begin{pmatrix} AM_{11} + AS_{11}M_{12} + GS_{12}M_{12} \\ AM_{21} + AS_{11}M_{22} + GS_{12}M_{22} \end{pmatrix} \tag{51}$$

If given that A, G are arbitrary, , this enforce the coefficients of A and G on both sides to be identical, hence we have the following relation:

$$\begin{cases} S_{21} = M_{11} + S_{11}M_{12} \\ S_{22} = S_{12}M_{12} \\ 0 = M_{21} + S_{11}M_{22} \\ 1 = S_{12}M_{22} \end{cases} \implies \begin{cases} M_{12} = \frac{S_{22}}{S_{12}} \\ M_{22} = \frac{1}{S_{12}} \end{cases} \tag{52}$$

The right hand relation is based on the second and the fourth relation Now, plugin to the first and the third relation, we get:

$$\begin{cases} M_{11} = S_{21} - S_{11}M_{12} = S_{21} - \frac{S_{11}S_{22}}{S_{12}} \\ M_{21} = -S_{11}M_{22} = -\frac{S_{11}}{S_{12}} \end{cases} \tag{53}$$

Hence, the whole transfer matrix is as follow:

$$\begin{pmatrix} S_{21} - \frac{S_{11}S_{22}}{S_{12}} & \frac{S_{22}}{S_{12}} \\ -\frac{S_{11}}{S_{12}} & \frac{1}{S_{12}} \end{pmatrix} \tag{54}$$

Conversely, the S-matrix in terms of the transfer matrix coefficients, we also have the following:

$$M_{22} = \frac{1}{S_{12}} \implies S_{12} = \frac{1}{M_{22}} \tag{55}$$

$$M_{21} = -\frac{S_{11}}{S_{12}} = -S_{11}M_{22} \implies S_{11} = -\frac{M_{21}}{M_{22}} \tag{56}$$

$$M_{12} = \frac{S_{22}}{S_{12}} = S_{22}M_{22} \implies S_{22} = \frac{M_{12}}{M_{22}} \tag{57}$$

$$M_{11} = S_{21} - \frac{S_{11}S_{22}}{S_{12}} = S_{21} + \frac{M_{21}M_{12}}{M_{22}} \implies S_{21} = \frac{M_{11}M_{22} - M_{21}M_{12}}{M_{22}} \tag{58}$$

Now, given that $R_l = |S_{11}|^2, T_l = |S_{21}|^2, R_r = |S_{22}|^2, T_r = |S_{12}|^2$, we have the following:

$$R_r = |S_{22}|^2 = \frac{|M_{12}|^2}{|M_{22}|^2} \tag{59}$$

$$T_l = |S_{21}|^2 = \frac{|M_{11}M_{22} - M_{21}M_{12}|^2}{|M_{22}|^2} \tag{60}$$

$$T_r = |S_{12}|^2 = \frac{1}{|M_{22}|^2} \tag{61}$$

$$R_l = |S_{11}|^2 = \frac{|M_{21}|^2}{|M_{22}|^2} \tag{62}$$

b. Let region 1 has amplitudes $\begin{pmatrix} A \\ B \end{pmatrix}$, region 2 has amplitudes $\begin{pmatrix} C \\ D \end{pmatrix}$, and region 3 has amplitudes $\begin{pmatrix} F \\ G \end{pmatrix}$. Also, let the first isolated piece (from region 1 to region 2) be described by transfer matrix $\mathcal{M}_1$, and the second isolated piece (from region 2 to region 3) be described by transfer matrix $\mathcal{M}_2$.

Then, the property of transfer matrices state the following:

$$\begin{pmatrix} C \\ D \end{pmatrix} = \mathcal{M}_1 \begin{pmatrix} A \\ B \end{pmatrix}, \quad \begin{pmatrix} F \\ G \end{pmatrix} = \mathcal{M}_2 \begin{pmatrix} C \\ D \end{pmatrix} \tag{63}$$

Hence, it implies $\begin{pmatrix} F \\ G \end{pmatrix} = \mathcal{M}_2\mathcal{M}_1 \begin{pmatrix} A \\ B \end{pmatrix}$. So, the total transfer matrix $\mathcal{M} = \mathcal{M}_2\mathcal{M}_1$.

c. Recall that for scattering states, the solution is given as follow:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Fe^{ikx} + Ge^{-ikx} & x > 0 \end{cases}, \quad \begin{cases} F + G = A + B \\ F - G = A(1 + 2i\beta) - B(1 - 2i\beta) \end{cases} \tag{64}$$

Where, $\beta = \frac{m\alpha}{\hbar^2 k}$, andthe second relation is provided by the continuity of ψ, and potential $V(x) = -\alpha\delta(x)$.

So, written in matrix equation, we get (from the relation on the right hand side):

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} F \\ G \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ (1+2i\beta) & -(1-2i\beta) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \tag{65}$$

$$\implies \begin{pmatrix} F \\ G \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ (1+2i\beta) & -(1-2i\beta) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 1+i\beta & i\beta \\ -i\beta & 1-i\beta \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \tag{66}$$

So, the transfer matrix is $\mathcal{M} = \begin{pmatrix} 1+i\beta & i\beta \\ -i\beta & 1-i\beta \end{pmatrix}$.

d. Since there are two parts of the delta square well, both with coefficient α (where $V(x) = -\alpha\delta(x+k)$ for $k = \pm a$). Then, since up to a shift of domain, both potentials are identical, they have the same transfer matrix. Hence, let $\begin{pmatrix} A \\ B \end{pmatrix}$ denotes the amplitudes on the left most region $x < -a$, and $\begin{pmatrix} F \\ G \end{pmatrix}$ denotes the amplitudes on the right most region $x > a$, the the method in part b implies that $\begin{pmatrix} F \\ G \end{pmatrix} = \mathcal{M}^2 \begin{pmatrix} A \\ B \end{pmatrix}$, where $\mathcal{M}$ is proided in part c. So, the transfer matrix is as follow:

$$\mathcal{M}^2 = \begin{pmatrix} 1+2i\beta & 2i\beta \\ -2i\beta & 1-2i\beta \end{pmatrix} \tag{67}$$

Finally, using the transmission coefficients derived in part a, we get:

$$R_r = \frac{|M_{12}|^2}{|M_{22}|^2} = \frac{(2\beta)^2}{1+(2\beta)^2} \tag{68}$$

$$T_l = \frac{|M_{11}M_{22} - M_{21}M_{12}|^2}{|M_{22}|^2} = \frac{1}{1+(2\beta)^2} \tag{69}$$

$$T_r = \frac{1}{|M_{22}|^2} = \frac{1}{1+(2\beta)^2} \tag{70}$$

$$R_l = \frac{|M_{21}|^2}{|M_{22}|^2} = \frac{(2\beta)^2}{1+(2\beta)^2} \tag{71}$$

$\square$

6

Question 6. *Linear Algebra Bootcamp I: Vector spaces (modified Griffiths A.3)*
1. Prove that the components of a vector in a given basis are unique.
2. Consider the set of all polynomials in x of degree $\leq N$ with complex coefficients. Does this set comprise a vector space, where the polynomials in x are the "vectors"? If so, suggest a convenient basis. What is the dimension of this basis? If not, where does it fall short?
3. Does the set of odd polynomials in x constitute a vector space? Why or why not?

Proof.

1. Let V be a $\mathbb{K}$-vector space (where $\mathbb{K}$ denotes the field). Let I denote an index set, where $\{v_i\}_{i\in I} \subset V$ is a basis of V. Which, the list is linearly independent, meaning that if $\sum_{i\in I} c_i v_i = 0$ (where each $c_i \in \mathbb{K}$), then it implies $c_i = 0$ for all $i \in I$.

   As a result, given any $v \in V$, since $\{v_i\}_{i\in I}$ spans V, $v = \sum_{i\in I} c_i v_i$ for some list of coefficients $\{c_i\}_{i\in I} \subset \mathbb{K}$. Now, for uniqueness suppose the two lists of coefficients $\{c_i\}_{i\in I}, \{d_i\}_{i\in I} \subset \mathbb{K}$ satisfies $v = \sum_{i\in I} c_i v_i = \sum_{i\in I} d_i v_i$, then $0 = v - v = \sum_{i\in I}(c_i - d_i)v_i$, with the claim of linear independence one must have $c_i - d_i = 0$, or $c_i = d_i$ for all $i \in I$.

   Hence, this shows that the component (i.e. the coefficient of each v_i for v) must be unique.

2. Let $\mathcal{P}_N(\mathbb{C})$ denotes all degree $\leq N$ polynomials with complex coefficients, to show it's a vector space it suffices to show given any $f, g \in \mathcal{P}_N(\mathbb{C})$, arbitrary linear combination $af + bg \in \mathcal{P}_N(\mathbb{C})$ (since choose $a, b = 0 \in \mathbb{C}$, this shows $0 \in \mathcal{P}_N(\mathbb{C})$, the set has a zero vector; choose $a, b = 1 \in \mathbb{C}$, it shows $f + g \in \mathcal{P}_N(\mathbb{C})$, which is closed under addition; also, choose $b = 0$ and arbitrary $a \in \mathbb{C}$, this shows $af \in \mathcal{P}_N(\mathbb{C})$, so it's closed under scalar multiplication).

   For any $f, g \in \mathcal{P}_N(\mathbb{C})$, there exists $\{a_j\}_{j=0}^N, \{b_j\}_{j=0}^N \subset \mathbb{C}$, such that $f(z) = \sum_{j=0}^N a_j z^j$, and $g(z) = \sum_{i=0}^N B_j z^j$. Which, given arbitrary $k, l \in \mathbb{C}$, we have the following:

   $$k \cdot f(z) + l \cdot g(z) = k \cdot \sum_{i=0}^N a_i z^i + l \cdot \sum_{j=0}^N b_j z^j = \sum_{j=0}^N (k \cdot a_j) z^j + \sum_{j=0}^N (l \cdot b_j) z^j = \sum_{j=0}^N (k \cdot a_j + l \cdot b_j) z^j \quad (72)$$

   Since each $j \in \{0, 1, ..., N\}$ satisfies $k \cdot a_j + l \cdot b_j \in \mathbb{C}$ (since $\mathbb{C}$ is a field), then $kf + lg$ is still a polynomial, and it has degree at most N (since it can only have nonzero coefficients for degree $\leq N$). Hence, $kf + lg \in \mathcal{P}_N(\mathbb{C})$, this shows that $\mathcal{P}_N(\mathbb{C})$ is a vector space.

   A convenient basis is $\{z^j\}_{j=0}^N$ (since a polynomial is 0 iff all the coefficients of each z^j is 0, so $\{z^j\}$ if contained in the space, is linearly-independent; and, since $\{z^j\}_{j=0}^N \subset \mathcal{P}_N(\mathbb{C})$, together with the fact that all polynomials is a linear combination of these, the given list does form a basis).

3. Let L denotes all odd polynomials in x (say over $\mathbb{K}$ a field), then any $f \in L$ satisfies $f(-x) = -f(x)$.

   Again, it suffices to show arbitrary finite linear combination still belongs to the space: Given any $f, g \in L$, and any scalar $a, b \in \mathbb{K}$, we have the following:

   $$(af + bg)(-x) = a \cdot f(-x) + b \cdot g(-x) = a \cdot (-1) \cdot f(x) + b \cdot (-1) \cdot g(x) = -(af + bg)(x) \quad (73)$$

   Hence, $af + bg \in L$, showing that L is closed under arbitrary linear combination, hence a vector space.

□

7

Question 7. *Linear Algebra Bootcamp II: Inner products (Griffiths A.5)*
Prove the Schwartz inequality $|\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle$.
Hint: Let $|\delta\rangle = |\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle$ *and use* $\langle\delta|\delta\rangle \geq 0$.

Proof. Given $|\delta\rangle = |\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}|\alpha\rangle$, we have the following inequality:

$$0 \leq \langle\delta|\delta\rangle = \langle\beta|\beta\rangle + \left|\frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\right|^2 \langle\alpha|\alpha\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\alpha|\alpha\rangle}\langle\beta|\alpha\rangle - \frac{\langle\alpha|\beta\rangle^*}{\langle\alpha|\alpha\rangle}\langle\alpha|\beta\rangle \tag{74}$$

$$\implies 0 \leq \langle\beta|\beta\rangle + \frac{|\langle\alpha|\beta\rangle|^2}{\langle\alpha|\alpha\rangle} - 2\frac{|\langle\alpha|\beta\rangle|^2}{\langle\alpha|\alpha\rangle} \tag{75}$$

$$\implies \frac{|\langle\alpha|\beta\rangle|^2}{\langle\alpha|\alpha\rangle} \leq \langle\beta|\beta\rangle \tag{76}$$

$$\implies |\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle \tag{77}$$

□

8

Question 8. *Linear Algebra Bootcamp III: Matrices (modified Griffiths A.8)*
Given the matrices $A = \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix}$
compute the following:

1. $A + B$
2. AB
3. $[A, B]$
4. A^T
5. A^*
6. $A^\dagger$
7. $\det(B)$

Proof.

1.

$$A + B = \begin{pmatrix} 4 & -1 & 2i \\ 3 & 1 & 2 \\ -i & 2+2i & 3 \end{pmatrix} \tag{78}$$

2.

$$AB = \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix}\begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} \tag{79}$$

3.

$$[A,B] = AB - BA = \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} - \begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix} \tag{80}$$

$$= \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} - \begin{pmatrix} 5 & -5 & 5i \\ 3 & 0 & 2 \\ 5-i & i & 5 \end{pmatrix} = \begin{pmatrix} -3 & 4+2i & -3i \\ 6+2i & 4 & 3i \\ -5-3i & 4+i & -1 \end{pmatrix} \tag{81}$$

4.

$$A^T = \begin{pmatrix} 1 & 3 & -2i \\ -1 & 0 & 2i \\ i & 2 & 2 \end{pmatrix} \tag{82}$$

5.

$$A^* = \begin{pmatrix} 1 & -1 & -i \\ 3 & 0 & 2 \\ 2i & -2i & 2 \end{pmatrix} \tag{83}$$

6.

$$A^\dagger = \begin{pmatrix} 1 & 3 & 2i \\ -1 & 0 & -2i \\ -i & 2 & 2 \end{pmatrix} \tag{84}$$

7.

$$\det(B) = -0 \cdot \det \begin{pmatrix} 0 & i \\ 2 & 1 \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 3 & i \\ i & 1 \end{pmatrix} - 0 \cdot \det \begin{pmatrix} 3 & 0 \\ i & 2 \end{pmatrix} = 4 \tag{85}$$

□

9

Question 9. *Linear Algebra Bootcamp IV: Eigenvectors & Eigenvalues (modified A.19, A.22, A.23)*

1. *Find the eigenvalues and eigenvectors of the matrix* $A = \begin{pmatrix} i & i \\ 0 & i \end{pmatrix}$ *Can this matrix be diagonalized?*
2. *Consider the matrix* $B = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix}$ *Is it normal? Is it diagonalizable?*
3. *Show that if two matrices commute in one basis, they commute in any basis.*

Proof.

1. Given A an upper-triangular matrix. By linear algebra theory, all its eigenvalues must appear on the diagonal (when it's in upper-triangular form), hence A only has eigenvalue i.

   Any of the eigenvector $v \in \mathbb{C}^2$ of A must satisfy $(A - i \cdot I)v = 0$, which $v = (x, y)$ must satisfy the following:

   $$0 = (A - i \cdot I)v = \left(\begin{pmatrix} i & i \\ 0 & i \end{pmatrix} - i \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} i \cdot y \\ 0 \end{pmatrix} \tag{86}$$

   Hence, $i \cdot y = 0$, implying $y = 0$. So, we must have $v = (x, 0)$, showing the eivenvectors of eigenvalue i are all of the form $v = (x, 0)$ (where $x \in \mathbb{C} \setminus \{0\} = \mathbb{C}^\times$).

   However, this matrix is not diagonalizable, since the only eigenspace of A is $E(i) = \operatorname{span}(1, 0) \neq \mathbb{C}^2$, so the eigenvectors don't span the whole $\mathbb{C}^2$, showing A is not diagonalizable.

2. Given $B = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix}$, its conjugate transpose (or adjoint operator / Hermitian conjugate under standard basis) is given by $B^* = \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix}$. Which, they satisfy the following:

   $$BB^* = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix} \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1-i \\ 1+i & 2 \end{pmatrix}, \quad B^*B = \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix} \begin{pmatrix} i & i \\ i & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix} \tag{87}$$

   Since $BB^* \neq B^*B$, then B is not normal.

   However, if consider its characteristic polynomial, it's given as follow:

   $$\det(B - \lambda I) = \det \begin{pmatrix} i - \lambda & i \\ i & -1 - \lambda \end{pmatrix} = (i - \lambda)(-1 - \lambda) - i^2 = \lambda^2 + (1 - i)\lambda + (1 - i) \tag{88}$$

   Which, if $\det(B - \lambda I) = 0$, since the discriminant of the above quadratic polynomial is $\Delta = (1-i)^2 - 4(1-i) = (-3-i)(1-i) \neq 0$, this indicates that $\det(B - \lambda I)$ has two distinct roots, hence B has two distinct eigenvalues.

   Since distinct eigenvalues must have linearly independent eigenvectors, with two distinct eigenvalues, the two corresponding eigenvectors must be linearly independent. And, with $\mathbb{C}^2$ being 2-dimension (over $\mathbb{C}$), this shows that there exists a basis formed by the eigenvectors of B, hence B is diagonalizable.

3. Given a finite-dimensional $\mathbb{K}$-vector space V (where $\mathbb{K}$ is a field), and have any linear operator $A, B : V \to V$. Let α be an ordered basis, and let $\mathcal{M}(A,\alpha), \mathcal{M}(B,\alpha)$ denote the matrix of A, B under ordered basis α. If $\mathcal{M}(A,\alpha)\mathcal{M}(B,\alpha) = \mathcal{M}(B,\alpha)\mathcal{M}(A,\alpha)$ (the matrix representative under basis α commutes), then given any other ordered basis β of V, let T_α^β denote the change of basis matrix from ordered basis β to α (which the inverse $(T_\alpha^\beta)^{-1}$ is the change of basis matrix from α to β), then we have the following:

$$\mathcal{M}(C,\beta) = (T_\alpha^\beta)^{-1}\mathcal{M}(C,\alpha)T_\alpha^\beta \tag{89}$$

Where $C : V \to V$ is arbitrary linear operator on V, and $\mathcal{M}(C,\beta)$ denotes its matrix representative in basis β.

Hence, we get the following:

$$\begin{aligned}
\mathcal{M}(A,\beta)\mathcal{M}(B,\beta) &= \left((T_\alpha^\beta)^{-1}\mathcal{M}(A,\alpha)T_\alpha^\beta\right)\left((T_\alpha^\beta)^{-1}\mathcal{M}(B,\alpha)T_\alpha^\beta\right) && (90)\\
&= (T_\alpha^\beta)^{-1}\mathcal{M}(A,\alpha)\mathcal{M}(B,\alpha)T_\alpha^\beta && (91)\\
&= (T_\alpha^\beta)^{-1}\mathcal{M}(B,\alpha)\mathcal{M}(A,\alpha)T_\alpha^\beta && (92)\\
&= \left((T_\alpha^\beta)^{-1}\mathcal{M}(B,\alpha)T_\alpha^\beta\right)\left((T_\alpha^\beta)^{-1}\mathcal{M}(A,\alpha)T_\alpha^\beta\right) && (93)\\
&= \mathcal{M}(B,\beta)\mathcal{M}(A,\beta) && (94)
\end{aligned}$$

Hence, A, B as linear operators still commute as matrices under basis β. Then, because β as a basis of V is arbitrary, A, B as linear operators have their matrices commute under any basis. (Which comes down to the fact that as linear operators A, B already commutes, so regardless of the matrix representative they should be the same).

$\square$